

RMOne AI Agents

Technical Architecture Documentation

Embed Integration System Specification

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Document Information

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1. System Overview

The RMOne AI Agents Embed System provides a secure, token-based authentication mechanism that enables external websites to integrate the AI-powered chatbot interface. The system is designed to operate within iframe contexts while maintaining strict security controls and domain-based access restrictions.

Core Components:

- Token Generation Service - Cryptographically secure identifier creation
- Domain Validation Layer - Origin verification and subdomain matching
- Role-Based Access Control - Granular permission management
- Session Isolation - Independent chat history per embed instance
- Token Store - In-memory authentication state management

2. Token Generation Architecture

2.1 Token Specification

Each embed link is assigned a unique 16-character alphanumeric token generated using the nanoid library with a cryptographically secure random number generator.

Token Properties:

- Length: 16 characters
- Character Set: A-Za-z0-9 (62 characters)
- Entropy: ~95 bits
- Collision Probability: < 1 in 10^{28}
- Generation Method: `crypto.getRandomValues()`

2.2 Token Generation Code

```
import { nanoid } from 'nanoid';

// Generate cryptographically secure 16-character token
const embedId = nanoid(16);

// Example output: "jgUSnbPDCMftMgPF"
// Stored in embed_links table as primary identifier
```

2.3 Token Storage

Tokens are stored in two locations:

- Database: Permanent storage in `embed_links` table with metadata
- Runtime Memory: Active tokens cached in `embedTokenStore` Map for fast validation

3. Authentication Flow

3.1 Embed Initialization Sequence

When an external website loads the embed iframe, the following authentication sequence occurs:

Step 1: Client Loads Iframe

Parent website embeds iframe with embed ID in URL path

Step 2: Origin Detection

Embed page detects parent window origin via document.referrer

Step 3: Token Validation

POST /api/embed/validate with embedId and parentOrigin

Step 4: Domain Verification

Server validates origin against allowed_domain in database

Step 5: Role Assignment

Server returns role (superadmin/admin/user) for the embed

Step 6: Session Creation

Client stores token in sessionStorage, creates embed session

3.2 Token Transmission

Unlike cookie-based authentication, embed tokens are transmitted via HTTP headers to bypass third-party cookie restrictions:

```
// Client-side: All API requests include token header
const response = await fetch('/api/query', {
  method: 'POST',
  headers: {
    'Content-Type': 'application/json',
    'X-Embed-Token': sessionStorage.getItem('embedToken')
  },
  body: JSON.stringify({ question: userQuery })
});
```

4. Domain Restriction Mechanism

4.1 Domain Validation Logic

The domain restriction system provides flexible yet secure access control. When validating a request, the following checks are performed:

```
// Domain validation algorithm
const requestDomain = parentOrigin.toLowerCase();
const allowedDomain = embedLink.allowed_domain.toLowerCase();

const isAllowed =
  allowedDomain === '*' || // Wildcard
  process.env.NODE_ENV === 'development' || // Dev mode
  !parentOrigin || // Direct access
  requestDomain === allowedDomain || // Exact match
  requestDomain.endsWith('.' + allowedDomain); // Subdomain
```

4.2 Matching Rules

Allowed: rmone.com != rmone.com (Exact Match)
Allowed: rmone.com != portal.rmone.com (Subdomain)
Allowed: rmone.com != api.portal.rmone.com (Nested Subdomain)
Denied: rmone.com != othersite.com (No Match)
Allowed: * != anysite.com (Wildcard)
Denied: portal.rmone.com != rmone.com (Parent Domain)

4.3 Subdomain Wildcard Behavior

Setting the allowed domain to a root domain automatically permits all subdomains through suffix matching:

```
// Subdomain matching implementation
const subdomainMatch = requestDomain.endsWith('.' + allowedDomain);

// Example: allowedDomain = "rmone.com"
// "portal.rmone.com".endsWith(".rmone.com") != true
// "staging.api.rmone.com".endsWith(".rmone.com") != true
// "rmone.com".endsWith(".rmone.com") != false (exact match used)
```

5. Role-Based Access Control

5.1 Role Hierarchy

The embed system implements a three-tier role hierarchy with progressively restrictive permissions:

SUPERADMIN

Full system access: All logs, FAQ management, chat editing, integration management

ADMIN

Elevated access: Own query logs, FAQ management, chat editing (no integration access)

USER

Basic access: Chat interface only, view own chat history, no administrative features

5.2 Permission Matrix

Chat with AI: Superadmin ' | Admin ' | User '
View Own Chat History: Superadmin ' | Admin ' | User '
Delete Own Chats: Superadmin ' | Admin ' | User '
View Query Logs: Superadmin (All) | Admin (Own) | User '
FAQ Sample Management: Superadmin ' | Admin ' | User '
Edit Chat Titles: Superadmin ' | Admin ' | User '
Integration Management: Superadmin ' | Admin ' | User '
Logs Tab in Results: Superadmin ' | Admin ' | User '

5.3 Role Check Implementation

```
// Server-side role validation
const embedData = embedTokenStore.get(embedToken);
const isAdmin = embedData.role === 'admin' || embedData.role === 'superadmin';
const isSuperadmin = embedData.role === 'superadmin';

// Client-side permission checks
const canViewLogsTab = isAdminOrAbove;
const canManageFAQ = isAdminOrAbove;
const canEditChat = isAdminOrAbove;
```

6. Session Management

6.1 Session Isolation

Each embed instance maintains an isolated session to prevent cross-contamination of chat data:

```
// Embed user identification
const embedUserId = 'embed_' + embedId; // e.g., "embed_jgUSnbPDCMftMgPF"

// Chat ownership
chat.userId = embedUserId; // All chats scoped to this embed

// Session storage (client-side)
sessionStorage.setItem('embedToken', embedId);
sessionStorage.setItem('embedRole', role);
sessionStorage.setItem('embedUserId', embedUserId);
```

6.2 Token Store Architecture

Active embed sessions are cached in memory for fast validation:

```
// In-memory token store (server/simpleAuth.ts)
export const embedTokenStore = new Map<string, {
  embedId: string;
  role: 'superadmin' | 'admin' | 'user';
  userId: string;
  validatedAt: Date;
}>();

// Token validation populates the store
embedTokenStore.set(embedToken, {
  embedId: embedToken,
  role: validatedRole,
  userId: 'embed_' + embedToken,
  validatedAt: new Date()
});
```

7. Security Considerations

7.1 Attack Mitigation

- Cross-Site Request Forgery: Token transmitted via custom header, not cookies
- Token Guessing: 95-bit entropy makes brute force infeasible
- Domain Spoofing: Origin validation via document.referrer and parentOrigin
- Session Hijacking: sessionStorage isolated per origin, cleared on tab close
- Privilege Escalation: Role stored server-side, not modifiable by client

7.2 Best Practices

- Use specific domains instead of wildcards (*) in production
- Create separate embed links for different roles
- Regularly audit active embed links and disable unused ones
- Monitor "last_used" timestamps to identify inactive embeds

8. API Reference

8.1 Embed Validation Endpoint

POST /api/embed/validate

Validates an embed token and returns authorization status with role assignment.

Request Body:

```
{
  "embedId": "jgUSnbPDCMftMgPF",
  "parentOrigin": "https://portal.rmone.com"
}
```

Success Response (200):

```
{
  "valid": true,
  "role": "admin",
  "name": "RMOne Portal - Admin Access"
}
```

Error Response (403):

```
{
  "valid": false,
  "error": "Domain not authorized for this embed link"
}
```

8.2 Embed Link Management

POST /api/embed-links

(Superadmin only)

// Create new embed link

```
{
  "name": "Client Portal - User Access",
  "allowed_domain": "client.rmone.com",
  "role": "user"
}
```

GET /api/embed-links

(Superadmin only) - Returns all embed links with usage statistics

DELETE /api/embed-links/:id

(Superadmin only) - Deactivates an embed link

9. Database Schema

9.1 Embed Links Table

```
-- MS SQL Server Schema
CREATE TABLE embed_links (
  id          VARCHAR(16) PRIMARY KEY,      -- nanoid token
  name        NVARCHAR(255) NOT NULL,       -- Display name
  allowed_domain NVARCHAR(255) NOT NULL,    -- Domain restriction
  role        VARCHAR(20) NOT NULL,        -- superadmin|admin|user
  is_active    BIT DEFAULT 1,              -- Enable/disable flag
  created_at   DATETIME2 DEFAULT GETDATE(),
  last_used    DATETIME2 NULL,             -- Usage tracking
  created_by   NVARCHAR(255) NULL          -- Creator identifier
);

-- Indexes for performance
CREATE INDEX idx_embed_active ON embed_links(is_active);
CREATE INDEX idx_embed_domain ON embed_links(allowed_domain);
```

9.2 Schema Relationships

The embed_links table operates independently but integrates with the chat system through the userId field:

```
-- Chat ownership for embed users
chats.user_id = 'embed_' + embed_links.id

-- Example: All chats for embed "jgUSnbPDCMftMgPF"
SELECT * FROM chats
WHERE user_id = 'embed_jgUSnbPDCMftMgPF';
```

9.3 Data Lifecycle

- Embed links persist until manually deleted
- Chat history retained per embed instance
- last_used updated on each successful validation
- Inactive embeds (is_active = 0) reject all requests

10. Deployment Configuration

10.1 Environment Variables

- APP_MSSQL_URL: Application database connection string
- CLIENT_MSSQL_URL: Client project database connection
- OPENAI_API_KEY: OpenAI API key for GPT-5 queries
- NODE_ENV: production | development
- PORT: Server port (default: 5000)

10.2 Production URLs

- Primary: <https://rmchat.vyaasai.com>
- Embed Pattern: <https://rmchat.vyaasai.com/embed/{embedId}>

10.3 CORS Configuration

The server accepts requests from any origin when accessed via embed tokens. Standard session-based authentication still requires same-origin or configured CORS.

10.4 Iframe Requirements

Parent websites embedding the chatbot must ensure:

- HTTPS protocol (required for secure context)
- No X-Frame-Options: DENY headers blocking iframes
- Allow scripts from rmchat.vyaasai.com domain

End of Document

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